

ActInf Livestream #029.2 ~ “Active Inferants: An Active Inference Framework for Ant Colony Behavior”

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welcome to the active inference lab live stream number 29.2 today is september 28 2021 and we will be discussing the paper active in france an active inference framework for ant colony behavior and today we are here with the first author daniel friedman the active inference lab is the participatory online lab that is communicating learning and practicing applied active inference you can find us at all of the links shown here this is a recorded and an archive live stream so please provide us with feedback so that we can improve upon our work all backgrounds and perspectives are welcome here and we will be following good video etiquette for live streams here at the short link you can find all of the upcoming and recent live streams that we have done so please get in touch if you would like to participate today our goal is to discuss the paper um that we just said active in france and we are going to um do some different papers in the upcoming weeks like i said get in touch if you want to participate so we're just going to introduce ourselves and go through a little bit of warming up i am blue knight i'm an independent research consultant from new mexico and i'll pass it to thanks blue for facilitating i'm daniel and i'm a researcher in california and i guess i'll start with what i'm excited about today i was just excited that in 2091 we could sort of forage out in a lot of different directions and make some initial trails to a few different ideas and then now in the 29.2 maybe we can develop some of those threads splash trails and maybe uh get some of it down in writing so it should look um pretty good nice yeah i'm excited too um there are some questions that i've been rolling over over the last week and so i'm kind of excited to delve into those and also into the code for the model today if we have time um being that it's just us i think we might get there well maybe though maybe more people will join there might be a little bit of lag i'm not sure so you might want to check that okay yeah um i don't know if it's my internet so the big question of this paper is um how can a group of active inference agents solve complex group foraging problems well first what i'll say is if you're recording it locally

as well as streaming then we can always upload a high quality recording and it does sort of bear on this big question this is a quote from the paper and that's sort of the big question in collective behavioral studies which is okay if greeting stephen if each agent if each subunit of the system had all the information needed well then there wouldn't necessarily be a challenge for coordination it's like if it already knew what to do then it would be a challenge but how do systems make adaptive decisions when subunits don't have all the information so in the case of an online team each person's only seeing a different part of the picture in the case the ants they're each in a little bit different part of their niche but those are some of the similarities which is that each nest mate is only going to be experiencing its local environment and we want to have colonies that work so welcome stephen maybe you can say hello while blue's just testing some stuff out oh great yeah good morning um well in north america anyway um yeah i'm based in toronto um and i work with communities and sort of systemic coaching approaches and very interested in how active influence scales in different ways so even though this is about small things it's also about lots of small things well your audio also just had some weirdness steve maybe talk directly into it oh okay is that better yeah oh okay but you mentioned the scaling and small the one one funny thought again this picture is from the ants that i was studying for five years in arizona um yes the nest mate looks like a tiny guy like a small ant but then we know that these colonies can live for 20 30 plus years they're also several meters under the ground and they weigh you know dozens of kilograms so then i'm thinking okay it's a similar age weight and height to myself and it's sub-use but let's see how's that is it any better it still is not doing anything i just reopened it but if it's not yeah then i guess okay now it's uh just the slides or just a jitsi so it looks like it's working now okay perfect yeah it's the slides open also it just is too much um yeah welcome back everybody fun times the show goes on foraging never stops so it does and steven will rejoin us hopefully he's having audio issues also it's a good day for technical uh mishaps we're having rainy weather in new mexico which is never ever happens so all the ants are underground hiding and the internet does not want to cooperate apparently this is so all right so speaking of multi-scale uh integration and across systems as happens in ants and in the ant colony um so we we talked about making this table last time and so here like what how is each system related to the agent the niche the interaction and stigma v so we started this this table do you want to uh maybe take us through what you've started here daniel yeah um see if you can [Music] get the slides again because it's not going to be visible at all on the stream oh okay i can also switch yeah i can switch the also yeah um so a lot of informal analogies and connections were being

drawn amongst different systems and that's kind of what complexity and systems thinking and active inference is about like finding patterns across systems so maybe there's other ways to break it up but at least one way we could at least start to classify and find similarities and differences between different systems would be this way with the columns so we have the system of interest like the ant colony as a system of interest then we can think about what is the agent level what is the sub unit level that might be the level where the active inference agent is being specified we then have the niche that the agent is operating in so that's the generative process that's presenting observations or outcomes to the agent which is embodying a generative model so that's related to this generative process versus generative model distinction the process is the the process actually outputs the like temperature or the observations whereas the generative model is a model of the process and then there's two kinds of interactions that are possible so there's sort of a transient interaction that might be amongst agents welcome back stephen there's the transient interactions amongst agents and in the case of the ants that's like an antennal contact and then there's the stigmatic interactions where there's a modification of the niche and so sometimes it helps to think about these interactions amongst agents that are um without memory as being like the transient horizontal interactions like the common example of the bird flock those are the birds looking at each other their interactions amongst agents that modify the likelihood of agents doing certain future behaviors but they don't leave a trace on the environment and that can be contrasted with or niche modification behaviors which are defined by being the ones that do leave a trace on the environment so maybe we'll explore different sections today but we can always say okay let's return to slide eight and let's add a row what's a system of interest let's add a column or let's fill in some columns so so far we have ant colony brain and internet slash computer abcs geb the systems that matter um steven what what would be um a fun system to throw into this mix and to look for parallelisms among we can't hear you steven yeah we still don't hear you steven yeah your microphone's definitely not working so i've started filling in for for brain and you know it all really is so dependent on what the what level you want to evaluate right so you know on one level there's like the little sub regions of the brain and you took us through i think in the dog zero video a very nice um example for ants that has been kind of done describing the different like organoids or substructures within the brain because definitely there are structures that all have different functions like the cerebellum is motor function primarily and you know the cortex is like higher level thought and and so there's you know these different sub regions of the brain but then also at the cellular level there are interactions

you know that that expand those regions that are from you know the cellular cellular level um and what is the so what the niches or the interaction um you know might might be different depending on like the the cells are responsible for driving the interactions between the organoids so i'm not sure what the level of um of the level of analysis the right one is and then stigmatizing modification i think about you know long-term potentiation um and like adaptation within the brain is very um maybe not stigmatic but it's a it's a really important modification i think um and sensitivity to different neurotransmitters and things like that blue it's still really lagging can you just stop streaming and i'm gonna pick it up on my side okay perfect go for it so stop the streaming and then just let me know when you did it let's see how this okay let's see what happens this could be very very interesting to do the live hack like this okay youtube we're gonna wait for it to catch up sorry everybody um so it goes okay wow we just hot swapped live stream feeds so we're back in the game no level of preparation that can prepare for laggy internet i suppose no and it's like half the time of mine and a half a time on your side but um if you're watching live definitely add any comments or questions because we're just going to be kind of exploring and chilling today but continue to um facilitate and i'll just be broadcaster role cool so the um yeah we were talking about the brain and the interaction and the sigmoidy and then for the internet i don't know what do you think is the the agent or what what is the sub unit of the internet like is it each server or where do you have the sub unit at daniel or stephen what do you guys think well so this this sort of ties into what they might say in in research you know unit of analysis as well but that's not necessarily quite the same thing as an agent um i i suppose on the internet ultimately it's the um it's it's whoever's um making who's that whoever's triggering choices in the system um which is normally a yeah it's normally a person interacting with the screen in some way i think the overwhelming amount of interactions on the internet are probably not human driven because any human is going to trigger truly a vast number of communications amongst other systems inside of the computer and probably remotely so maybe and i think this is one of the unique uh challenges but also interesting areas is like the internet is a cyber physical system so we could have one model where the agent like the subunit are physical devices so we kind of draw the blanket we um focus on the partitioning around a physical object like internet of things we have partitionings around things we could also have a human-centered partitioning so partitioning around humans but then there's a lot of gray space in between the humans a lot of links that aren't human between the human links and then another approach might be purely informational or computational and seeing the agents as software agents but i think that

interface where how will we talk about the software and the hardware agents coming together is where the internet of things is developing and what it's coming to mean i mean one thing you could think about there also is when say for instance i make a choice or i do something on the internet and there's a whole load of there's a cascade of other things going on through the different routers and the servers etc is that more like the pheromone trail that sort of dissipates off into the distance leaving records and how much of its like other agents initiating because often you know every time you do something to some extent yeah databases might update however and this may change to what extent that is alive is another question you know is it is it sort of a computational form of niche modification that's going on in most cases in terms of the internet but what they're trying to do because it's that idea of the battle for your mind they're trying to make it so that in most cases that the user has less has less thinking to do in terms of the choices they make and the design there's this there's a great book in web design called don't make me think and the idea is that you go somewhere it's almost obvious what you need to do a bit like when you pick up a newspaper you know that you're just going to turn the pages how much of all this architecture in the background is to support giving you what you expect to be given when you expect to be given it um and how much of it is um which you know how much of it is actually um what you might term an agent with with some sort of teleology well you mentioned expectation there and um expectation is something that agents engage in so maybe it's even we could keep on forking this off but what do we know from the active inference ontology what defines an agent external states blanket states internal generative model what does the internal generative model entail well it includes predictions about external states it includes policy affordances it includes preferences and includes expectations over some of these things as well so when we say agents we can be referring to the whole markov decision process or other ways of viewing the active agents and then the expectations of the agent are really important because it's not just that psychologically it's going to determine whether the agent likes it or not but it helps us connect these systems like the ant colony and subunits of the brain so we don't need to use only the psychological interpretation but then we can have a psychological interpretation where one seems warranted given the system of interest but also we're going to be able to compare that and look for similar functional features in systems without getting into just some sort of abstract debate over whether it's alive or whether it's certain feels a certain way to be that system so daniel you brought up the generative model and i was just going to bring it up um right before this and so i've been really uh you know thinking a lot about the generative model and like where

is the generative model which is like i think probably my favorite podcast episode that i've ever cut is going to come out next week and it talks about like the location of the generative model and so you know it doesn't necessarily have to have a location because it's like software right like so i mean that yeah there's like a copy of the software running on disk maybe maybe it's in the cloud like it does but it doesn't matter because it's the program that's running so the location of the program is almost not even relevant um so i just i think about like what is the generative model of the internet right like what does that look like great question um inactivism for cyber physical systems and embodiment uh in encodement maybe there's some analogous term to what would describe what we call 4e in the embodied extended and cultured etc for digital systems because they do have important attributes that in a strange way almost suggest that some of these attributes some of these features that are suggested to be quintessentially related to having a body and embodiment also exist for software could i putting something on to that is say we think about rather than and you've got the active inference generative model but say you've got an active niche modification algorithm so ie it's still off the agents being making choices but what's happening is so for instance say you've got abc testing happening on a website so basically they send out three different campaigns with three different types of profiles now the user makes choices based on how the user makes a choice they may be across the whole of the a whole market the abc choices might get modified in that niche but eventually maybe then actual clients get sent down certain routes um i mean a classic case of this is also what happens with those in-store cards because i actually i ended up going i didn't get go for it again i had an interview with the nectar card people in the uk once and they uh they do these cards that you get in store and basically i think it worked out they have like a million permutations of offer that they can give you based on what your different buy-in options are so in a way they're kind of that but they only serve you up then whatever you see one of those right you don't see what all the other people are getting and some of them are small changes some of them might be bigger changes so in some ways it's like it's still the it's still like it's like the niche is modifying to the agent's adaptive behavior but the actual niche isn't necessarily doing adaptive active inference but there may be some sort of active inference in a more kind of i don't know non-living way there might be um we'll probably have better words and ontologically informed ways of talking about it but there's kind of like low bar active inference or like austere active inference that is just any input output structure it's going to be able to be fit into something like a partially observable markov decision process so whatever it is that the niche is doing it's emitting observations so we can model it under this

certain low bar active inference versus the uh sort of anti-dissipative truly cognitive strategic active inference and um also there's no reason why we can't have hybrid models where there's like active inference agents in a niche that isn't an active inference agent so like in this ant model the niche is just a landscape of pheromone densities with a random dissipation factor external niche could be made very um much more complicated and the observations that it passes back to the agents could be like customized to them and that could be calculated by an active inference niche or not and that's sort of the extremely interesting work of axel constant and others with some type of symmetry occurring with the agent and the niche jointly inferring each other so like what is the niche's generative model of the organism what is the internet's generative model of the user so it's interesting questions i love that idea of symmetry especially between the agent and the niche um and its relationship or potential relationship to the kind of gaia theory that the the niche has to expect or have an expectation of the organism that's going to interact with it or create it or modify it or exploit it in some way um in order for there to be some kind of reciprocal um interaction there so uh super nice um maybe we can come back to looking at this um table and maybe think of other systems that might be interesting to go through and move on to some of the um questions for 29.2 but wait before we go to this one i want to go to this one hold on ha so we were talking about um the role of dopamine last time which slide uh so sorry this is slide um i skip to i'm sorry i can't see i think it's number seven yeah number number eleven oh wait okay so um so we were talking about last time the role of dopamine and like influencing the affordances and and this really kind of speaks to the role of reward in a generative model and and so you know we talk about one of the beautiful things about active inference is it kind changes the reward um structure in the model because you know there's some kind of reward that's given to you know actually finding the food right so so there's some some reward for finding the food but also and and even in this model so the the reward structures has changed right so there's um it's not just about finding the food um and maybe this is a good opportunity to to look through the code maybe as well but but you know we think about the change in the reward structure and we place some value on the epistemic um you know the knowledge about the model of the world so if we can build a more precise model of the world we can also um gain more food overall so i don't know if uh you guys have comments on that or yes steven guide yeah i think also as well as built like you say building that model of the world becomes a it's an action model okay so it can be what might be seen factually spurious but practically useful you know so it may be really useful to have some bizarre ritual about how to get on a soccer pitch and try and kick

the ball right but if it does something better than what you're doing otherwise go for it and i think the challenge comes with um increasingly esoteric rituals if they're not related to the actual causal structure of the world if they work for a moment they're not going to be adaptive because if your only strategy is you know praying to the kicking god instead of practicing it might work or it might indirectly induce some sort of outcome but is that a long-term strategy um but agree that the generative model is including a model of action it's not just about getting precision about how things are out there because what ultimately matters is getting precision on our actions and then there's sort of these two parallel paths or two intertwining paths which is do we take a reward centered approach is behavior dominated by reward and then the other side of the reward coin is like the um punishment aversion so carrot in the stick that's sort of a sister road two-lane road do we pursue reward or do we avoid regret and suffering and then a third road is this uncertainty minimization approach so i pasted in this paper by columbo explanatory pluralism they review three prominent hypotheses of dopamine activity so anhedonia which is unpleasurable states incentive salience and reward prediction errors and i think this is a really fascinating paper and it does apply to the ant context because they make a very specific system point talking about the role of dopamine in the mammalian brain reviewing empirical evidence and then also pull back to the broader implications for the philosophy of science they conclude that the evidence currently vindicates explanatory pluralism in other words these explanations do not subsume or knock out each other there's a pluralism there's multiple explanations that exist that capture different parts of how the system really works the vindication implies that grand unified incl uh claims of the advocates of the ptb are unwarranted look at what that is again but more generally we suggest that the form of scientific progress in the cognitive sciences is unlikely to be a single overarching grand unified theory little possible indirect reference to those who claim that the value or the utility of for example the free energy principle or active inference is just simply that it explains everything i don't need to even say any more or detail where i used it or show you the code or show you that it's better than another model the fact that this model i can fit everything into is the utility and it will subsume other models and other frameworks now that's a contentious point that has to be either empirically supported or not but it just is so interesting how dopamine is reward molecule or the other side of that coin with the punishment or as a precision molecule the empirical evidence alone are not enough to disambiguate that and so you know trust the data let the data speak but then this is just showing us a kind of real time that our models of the world get projected as we do sense making on the experimental outcomes and um it's not

always as clear as it seems well you know it just reminds me like so i mean just from a very literal sense if we think about dopamine as um a reward for minimizing prediction error like how much do we all like to be right all of us so i mean like just just for being correct like that in and of itself is a huge like dopamine reward and i think about like i'm a super ultra nerd and so i geek out on things like difficult math problems or solving a rubik's cube or or these kinds of things and i'm just like finally when i get or even code right like finally when the code runs there's like this huge like something that you've been working on for a really long time it's like a huge like dopamine rush for me which is why like i keep going back to doing these kinds of things but but ultimately that's a prediction error reward so it's interesting to think about and here's sort of the two ways that they're phrased and it's somewhat subtle but i think it gets at the difference between reinforcement learning and active inference is there a drive for being right and then we prefer to be certain or is the drive for certainty and then there's a preference over being right or being accurate so are there situations where it actually brings us solace even though the direction of the outcome isn't what we expected or isn't what on some sense conversation is what we prefer but actually the fact that we're able to have certainty outweighs the fact that the outcome went the way it went so and maybe there's other phrasings too but it's just again which one do we put in the driver's seat precision or reward well so what's the difference between being certain and being correct like being right and being certain like okay if i'm really certain but then that that certainty is unwarranted then that's when i'm wrong but then i'll i have this prediction error gap right so either way so what's the difference really between being certain and being correct well i think one if i could just jump in there i think one thing is really interesting is if everything's based on energy minimization like you've got you've got a process you're going to keep doing it and it's optimizing to make it basically if i use less energy i'm being more efficient i'm being faster i've been smoother so it always seems like yeah go for the reward the thing is though once you get into something where it's more complex or uncertain you know the idea and they talk about this now actually there's a whole area in um development work and design thinking called failing forward fail forward so the idea is fail fast fail quick but part of this is like in you know and i always say within ethical bounds right because it might be great for the designer as long as it's not a community member being sort of being frustrated by this but but it makes sense in the sense that and i don't think reinforcement learning can really account for this is the benefit of doing something because it might be something which doesn't work but now you can take from it it's almost like you're giving yourself a sample of the counter factuals in a way by something going wrong but you

don't obviously we can then break it apart and say well there might be something right still in that you know and some bits which are more wrong and we can sort of do stuff with that information almost like succeeding forward would be better than failing forward i mean failing forward has this psychological benefit of helping people recognize that it's okay to not get it right the first time and just get something on the paper and we'll iterate from there so the rhetoric of fail forward makes sense but having something that's even positive like succeed forward or win forward or contribute forward or learn forward or develop forward those actually capture the functional spirit that um is here and let's think about how that relates to blue's question so what's the difference between being certain and so there could probably be many takes on this but i thought of this classic science textbook you know when when they give you the science textbook um the difference between accuracy and precision now in many cases now this also is not coming from the active inference ontology so they may be using these terms in a slightly different way but the point is we're getting at two complementary components of precision basically uh we can see that there's these two off diagonals the top left and the bottom right so high accuracy high precision it's like your average is on target so that's the accuracy is where you're aiming at accuracy aim and then the precision is referring to the tightness of the distribution around that mean and so you could have high accuracy like it's always at the bullseye but the precision is low or the uncertainty the spread is high you could also have like a thermometer with it giving you very tight readings high precision but it's systematically biased in a certain direction this is just one way to get at two different aspects now we've seen this in a few different cases in variational inference as being the um difference between the expectations the means of the distributions that's like the accuracy and then the precision is like the variance of that distribution so you could have a very tight distribution um the the q distribution very tight and close to the p distribution that would be accurate and precise you could also have a really over fit over confident model that is just simply quite divergent from the underlying p so um which one is being optimized are we trying to aim for a tight scatter and then we and then second order is we want it to be in the middle or do we aim for the middle and then the second order is we want them to be accurate stephen yeah that that to that question is a good one um and i think distinguish because often they might get conflated in many ways but it's good to pull apart precision and accuracy because it is a you know you could have um you know you you could be very precise in the fuzzy set that you have and you could but it could still be fuzzy you could have something like you say more accurate but it moves around a lot and i think that that's that's an important distinction that

might be lost in many cases yes and it's actually one reason why it's important to sort of stick with the uncertainty and pass the estimate the state estimate as well as like the uncertainty around a state estimate pass that through because if you collapse the uncertainty and you treat all estimates as being equally precise there's this whole dimension of the data that has been dropped for example in gene expression analysis sometimes there's uncertainty over whether a gene is differentially expressed or not like it has a distribution and you pick some cut off but then sometimes people take the genes on one side of that line and then they just move forward with them like these are the differentially expressed genes as if it was a discrete categorization now take a continuum and then discretize it which makes certain kinds of analyses really simple or straightforward but a whole aspect of the variability of the data has been completely crunched out yeah and this also then ties into scale as well um you know you could and and how much so like you mentioned the thermometer so the thermometer gives a fairly um well fairly precise sense of the of of the reading it's fairly consistent and um it's averaging out the average is is fairly relatively accurate in terms of the average but it's not going to give you the range or the the spread within that and that that can apply to a number of things and then you as you go down to less and less if you for instance i suppose like you would say if someone's purely looking at one ant or one ants behavior at a time then you you might have this sense of i'm being very accurate in terms of my ant analysis so to speak however at the level of the you know the larger colony you're you're you're not as accurate you know so there's some interesting sort of scale things that can't be avoided really yeah are ants accurate or and or are ants precise the nest mates it's almost like maybe we could think of them as being um high accuracy but low precision like let's just imagine what that would be like for the colony so each forager is in the neighborhood of making adequate foraging decisions but there's a lot of noise around that so over thousands of interactions and foraging trips you're going to end up getting a lot of your points in the middle so it's cheap because it's lower precision so you can have simpler subunits but it's centered around the target whereas if you have it centered over something that's not the target it's going to be sub optimally adaptive for the colony but i think ants just show that um it's not about making every single action perfectly the right one it's about being right more often than not and then just doing it many many many times which is related to that sort of um develop slash fails succeed forward sorry go ahead steven just just one thing i think this is also useful just in the sense of and it's almost implicit with the agent-based modeling approaches that get used in biology but it's quite interesting in the sense that that brings in that inter-subjective piece what are these ants trying to do for instance how are

they trying to organize what's going on rather than the first person perspective i what do i see what's my perspective on the ant as an observer which might be more psychological in our world or what is it that i see the group doing on average like the temperature but when you start to get into agent-based approaches um it it does allow this second person um stuff to happen which is really i know in psychology is the missing piece really in a lot of cases so um you know how this how this in a way also ties into thinking about agents rather than thinking about what i see someone or something doing from the outside and inferring from their behaviors externally qualitatively sets up a different type of um experiment or modelling so i wonder about the different um scales like as we scale from the nest me to the colony like are are they comparable like are they asking the same question like the colony wants to know like will there be enough food and the forager never asked that question the forager always wants to know where is the food right like so so they're asking very different kinds of because the forager knows it has no uncertainty that that its job is to find food right and and so the colony wants to have enough food but the forager always wants to get the food and i wonder how this kind of asking different scales to things like the brain and components of the brain like or the internet and components of the internet like like what are our is is what defines a scale whether we're asking the the same question or not like or or is it the goal of the system that defines the scale of the system i don't know it just made me think about yeah i mean that's a good that's a really good point i mean yeah you could if you're starting to look from the agent and you could even think about that in the brain like yeah if different parts of the brain are foraging information so it's not like that that would explain a little bit why you have these kind of parts you know parts the you know different parts of the brain have different personality um queries going on even when certain parts of the brain get sort of put to sleep so to speak so rather than yeah i mean that could be that could explain and if they're if they're each working away i mean like a colony um and yeah who which one is in some of them get to be in the driving seat and some just yeah some get to ask different questions or some get to sort of have their questions to influence other questions and some of them just get on and do their thing if that makes sense you're a forager basically you're gonna forage right that's the deal but someone else it might be like i'm hungry you got a forage more right oh just i i rapidly assembling some image memes it actually breaks down well into the the chess and other fields other serious games of tactics and strategy um strategy so like blue your question the colony was asking do we slash i we don't know what pronouns the colony uses do we slash i have enough food it's as if it's asking that question does it identify as a entity an i or does it identify

as an ensemble we don't know but then the nest may is asking the more tactical and action-oriented question which at a first pass is like where is the food but we know that in active inference what matters isn't just like simply inference on the location of the food out there but what matters is inference on our action selection for how we're going to get that food or how we're going to implement a strategy that has a good trade-off of us finding it so like here chess meme tactics mean doing what you can with what you have so like whatever pieces are on the board you're gonna have to do some sort of action selection for your next move here strategy requires thought tactics require observation so it's kind of interesting because observation is one of our terms in the ontology like the agent is receiving observation without observations tactics cannot be implemented and then strategies requiring mental action just like our previous discussions in 28 the thought is actually the planning over observation strategies uh nice double blue and um there's other definitions of strategy and tactics and how they're similar or different so that's just kind of like cool that um it's a way to think about maybe not just that are the levels asking different or the same questions but might levels be defined or delineated by the difference in the questions that they ask independent of the spatial size of the subunit the more tactical is like more agent level and then the more strategic requiring reflection or cumulative culture or mental action that level is being implemented by higher and higher nestings wait blew your mute because you went down to just one can you hear me yeah perfect um i i went down to just one so you can share the names with us because i'm missing out on the meme action and i like i can't stand it a little bit i'm so sorry so so speaking of action and and different levels so this is something that i've really like been thinking about over the course of the last week uh this is like the next question on on on this slide 11. is this this prospect of intergenerational stigmergy and i just wonder like you know as we were both foraging for memes and slide content like creating the 29.0 like we have this internet right and so like we just google search for whatever and informational foraging i know is not the same as maybe ants foraging for you know a food but um you know we're both foraging we come across the same kind of content and ended up like putting it into the 29.0 in different ways in different places and and i just wonder you know there's this concept of of intergenerational stigmergy that's very uh i and i wonder if this is human-centric or if there are other examples of this in the animal kingdom and perhaps among ants i know daniel we've talked at length about the different foraging habits for example of different um different foraging habits of different species or the same species of different colonies and so where do they get this habit like to forage differently like at different times of the day or whatever and and it spans

generations and so is there some trace left in the environment like humans we leave traces for the next generation i mean we directly do some cumulative cultural evolution transfer um and there are other species i know that that teach like dolphins i know teach like some foraging habits and so forth so so what other examples do we have of this kind of intergenerational stigmergy or like you know we leave traces like the time capsule of you know like what to find like every decade we open up the time capsule from you know whatever the previous decade was and find things like eight tracks and like whatever things that are put in the time capsule that are just like oh look way back in the day um so i don't know what do you think about that um just relative to ants and relative to this intergenerational stigma g and the different foraging habits of colonies from the same species good uh where did the priors for foraging come from and then to what extent is there scaffolding of one nest mate in their foraging activity by older nest mates which is actually within a generation that's like sort of some skin cells being scaffolded by previous skin cells it's like if you just kill all the foragers of a colony the foragers that step into those rules will perform sub optimally that's actually related to colony collapse in the honeybees so there's some element of like they have an inkling of how to adapt their foraging yet um also there's flexibility um i mean it just makes me think of how we could actually study it like putting nest mates in different environments or having uh one colony and splitting it and training it a certain way and then reintegrating them and seeing if what mattered more was like the environment at that generation or whether it was possible to have epigenomic or genomic changes that sort of lock in and and uh shape foraging and then on um 12 you know i'm doing a little live meme modification like foraging is not an act but a habit with a probably misattributed aristotle quote so it's almost like forging is not an act it's a policy it's a policy over specific motor actions but yeah the intergenerational part is very interesting and it i'm just thinking if there's other true intergenerational ant scaffolding one example not in foraging is like when a leaf cutter ant queen goes on her nuptial flight she takes a little sub-sample of the fungal garden and uses that to start the next generation which actually i think relates to the second point you had about 30. so that's super interesting and also um directly related to um the microbe transfer right like you think about um microbial transfer that happens um like through the birth canal like there's a difference between c-sections delivered babies and babies that actually pass through the birth canal just in terms of how they're inoculated as well as breastfed babies versus formula fed babies there's a huge like inoculation difference with your like but just microbial flora and so that that is the uh and i wonder what um you know this speaks to the holobiont concept like we're not just

humans we're humans inoculated with all kinds of things and maybe ants aren't just ants they're ants plus that fungal and i have there's been studies on that fungus like on the fungal composition is it different from colony to colony um do you know i don't know i know that uh one group in my previous lab they tr they isolated nest mates and then they treated them with antibiotics and changed their microbiome and they and others have also shown that there's a role for the microbiome in recognition because it modifies the smell of the ants that modifies how they recognize each other um maybe that's more like a dialect like that's kind of how they recognize that they're in the very short time range like in the same group or they're they're likely to be collaborative but i i returned to eight to think about what you just described about like the body and the inheritance microbial forms like is there a way to refer to that human is the species but then you know if humans contain and are multitudes you know there's the body there's the other life forms there's social uh interactors humans aren't the species but the host which ties in very nicely to our next um live stream which i'm very excited about which um with matthew sims he's gonna come and talk to us about uh the symbiosis between squid and vibrio fischeri in the light organ which is something near and dear to my heart i worked in new mexico i don't know uh how is this possible in um right upstairs you know you have those labs that like you know everybody in the lab and you guys can come and work and um like share equipment sometimes and so i knew this lab and our lab were very close uh the lab of michelle nishiguchi and she studied vibrio fisheries and the light organ so i knew all these people doing directly related projects but that's it's a really cool uh concept that you know maybe we're all in some form of symbiosis and i know in mice they have sterilized mice and the mice grow up totally neurotic like mice lacking the microbiome free mice like they're completely neurotic and have way abnormal behavior and they've linked a lot of you know human behavior or human abnormal or abnormal behavior or neuroatypical behavior to microbiome stuff as well so so it's really interesting to think about maybe we're all symbians in some fashion i mean and like maybe we're not the species but we're just the host reminds me of um demetrius bolus speaking about the through others we become ourselves and it was situating active inference in the dialectical tradition and um you have a coming together of different pieces and they don't reduce their distinction they actually become especially over evolutionary time more polarized like the nuclear genome becomes more nuclear the mitochondrial genome becomes more reduced in mitochondrial or the host becomes on one trajectory and the parasite goes in the other they don't converge and equalize each other they end up becoming increasingly specialized parts a newly emergent type of system and then these multi-specified systems can

be fragile because maybe now there's a failure mode of the part and then that causes the whole system to fail so like bucky fuller would always talk about how um all cases of biological extinction are due to over-specialization might be a bit of an overstatement but that's sort of the idea is like generalists can deal with changing niches better because they're always able to hedge their bets amidst uncertainty and average out over all the different um foraging investments whereas a specialist does really well until they don't but in life it's not just like you can do well most the time and then you can fail sometimes living systems need to be succeeding at least adequately all the time if any part of your body was under producing atp for like five minutes it would get tissue damage so it's better to be adequate all the time rather than hyper successful for even almost all the time and then what kinds of strategies are resilient amongst that kind of uncertainty and situation so about halfway through what you were saying i wondered when we started having a political discussion because it just really you know speaks specialization or maybe polarization um which we see in society so like you know we have these different factions or different like sub types or sub classes of of people and biological systems that don't become more like one another they become more separate from one another and i think um you know in the resounding echo chamber that is um a self similar political environment or class environment you know it becomes this this echo chamber that that um you know just increases our certainty right increases the the validity of our own model and it makes us question the model of others so i wonder uh and this brings us back to uh what we're going to discuss with matthew sims is that right or is the jet like is the becoming more similar for the fisheries vapor fisheries in the squid or is it becoming more different i i just wonder yep uh um so through interaction do we become more similar or different or what kinds of systems through interaction become more similar or different and then how can systems be more anti-fragile given that relationship amongst the subunits versus which which failure modes are you de-risking on and which failure modes are you increasing the risk on so it's an interesting question i mean it's not just political it is very um broad question blue let's look at the code and maybe just scan through some sections of it and then start to see um in on 11 before we get to the code it's just like what else could we include in an ant model like this was um we talked about the natural history of the ant that we templated off of the formica species in the paper but um we used one pheromone we didn't have interactions you know we had such um a sparse pulling over of features of ants into our model so that's why it's just like a introductory framework so what what other pieces could exist for ants and we can kind of use the active inference um breakdown of what perceptions could we add in

tactile interactions other olfactory interactions light intensity the color the polarization of another type of polarization what type of cognition could we add in so this was like a one-step decision-making ants it just makes a decision for the next time step based upon the local pheromone density what kind of cognitive processes actually exist in ants or what kind might be interesting to see in different systems like memory and then what actions exist here we had a movement action affordance any of the directions and then we had sort of a pheromone deposit affordance that didn't actually have a specific um control policy it was just a if then if you already touch the seed then you deposit pheromone if you're still looking for the seed then you don't deposit fairmont so we can think what are the perceptions cognitions and actions of the aliens as currently modeled in the paper and then where can we go with that to go from that question about the simulation to meet the empirical data the other side of the table what kinds of ant data already exist there are several very fascinating databases that include genomics gene expression transcriptomic and epigenome or epigenomics datasets they're sparser for ants than they are for humans and mice but they are out there and there will be more of them in the future what kind of behavioral data already exists ethology the study of animal behavior video data maybe they analyzed it under a certain framework but now it could be readdressed and then like rfid and qr codes that can do semi-automated high-throughput tracking so people attach rfid chips to honeybees and track their foraging activity and then we also have information on their ecology and morphology so features of which we're not in our simulation so this is just this sort of prime looking at the code by asking like how could the code be modified and then how could the world already be seen as reaching out to us with these helpful pieces of data and then where are we going to be in the middle like what kind of model is going to connect the empirical data sets that do or could exist to the models that do or could exist and then what questions are most important because we want to be asking relevant questions and so it's better to be keeping the goals in mind as we look through everything okay let me just re-crop this window while i get the code up anything you would just say on the overview or like general thoughts on the kind of multi-agent simulations or foraging simulations that you've made uh just i have some questions about this simulation actually or some some things that i would think about to include as parameters like in a dream simulation where we you know could can simulate whatever with unlimited computational power and relative world data um so i think that here i'm interested in nestmate interactions so i think that that would be something that would be interesting to model like the different types of ants that exist like the nurse maids and the foragers and the queen and the uh reproducers i can't

remember we went through all the different types of nest meats last time but i would just be interested in in kind of simulating how as a colony ants interact and function i know that that's like you know very whatever big and i mean in most simulations that i've always done it's like the same type of um the same type of agent right like so you give the agent different options but it's never like this agent doesn't forage and this agent does i don't know but i mean i know that they do have different people do do some modeling with different types of agents but i just wonder what the compute power is and what the validity of that is yes great we're gonna um take a sort of uh a stepwise approach and one thing that i think will be fun there's people who know way more about python and coding than both of us for sure including alec chance and others and then there might be others for whom they've learned about active inference or they've read a lot about it maybe even looked at the math but here is going to be where we see how it looks in code so there are these multiple languages and ways of communicating and learning by doing active inference like natural language math as a language programming as a language art as a language maybe there's other ways too but at the very least we know that we have natural language mathematical formalisms like the equations that we're always putting up and then here's the code those threads might be looking more similar or different as we start to learn more about like interactive notebooks and ways to integrate our language use of the natural kind with the programming kind we're gonna do kind of a a step down first we're going to look at the config file so the readme file in this case is not super informative which is fine um but the config file is where we're going to start that's going to tell us about like what kinds of variables and parameters matter then we're gonna look at figure one and that's just sort of gonna be like okay this is how you run it and then we're gonna pull back to the ans dot py and look at what functions and subroutines are actually defined inside the code so inside of the config file it's only 30 lines long 31 lines long and it has some pretty clearly defined variables that are being set and the idea is that all the parameters are stored in this file so that you can kind of hit play and then it will look to this config file we'll see where that happens in a second the first block relates to um if you want to add the ants in a staggered fashion how many how many time steps should happen between adding new ants in uh how many or what's the location of where the ants should begin nest factor we can look at i'm not exactly sure it is here it relate i'm not exactly sure where what the nest factor has to do it might have to do with the distance of the nest let's ask alec on that one the grid is the side size of the map so here's a 40 by 40. and then here is a very interesting block it's the focal area which is three by three so that's the grid around each nest mate is just three cells by three cells and so this could be

changed to you know like four by four or something like what happens when the ant has a broader range of perception than it might have for action and that's the action map and so those are the um the actions that can be undertaken relating to basically moving like down into the left or up into the right the food location and the size of the food are provided then the wall so that specifies like within the grid what is the shape of the maze inside of the map 24 and 25 describe the the number of discretized levels of pheromone that exist and then a factor that is the stochastic decay of the pheromone trail here the observation states so like the α in the active inference model defined as the number of pheromone levels so in this case those are both 1 or 10 discrete models it's so interesting because it opens up the door for the generative process to differ from the generative model so like the observations could be there could be um three categories of observation low medium and high and so you could still have a continuous variable or discretized variable in the world in the generative process but the generative model could be over a reduced version of that which might be all you need to know maybe there's enough stochasticity such that just knowing if you're in high medium low tells you whether to put on a jacket or not you know you don't need five decimal points on the temperature and um the number of action states is also reduced to the focal size but just as mentioned that could also diverge and then the um the length of the simulation i think is specified in 31. so those are some of the parameters that exist just in the base simulation some of the degrees of freedom that do exist and that's even before introducing these like other sorts of um pieces that could exist okay any thoughts here before we go to figure one just like you know i mean i i always want to know like why like why did you choose this or like did you play with the decay factor or did you play with like the length the time steps i'm sure like maybe your reviewers commented that as well but um i just wanted like i'm always curious to see like was there any unexpected outcomes or optimal outcomes that occurred like peak like did you pick 500 time steps because that was you know best or the decay factor of 0.01 because that was most real realistic or those types of things good questions and i think we we probably could have um organized these with some sub headers and some comments a little bit more clearly we're going to look at how the simulation gets specified and then pull back to like what are the subroutines so the the top part of a python program is basically pulling in and then giving nicknames to other resources that are required so these resources can include packages like numpy which is a common toolkit used for math and statistics in python i'm sure blue can say more on that one and then there's this um the config as the `cf` so that's when we call like `cf.dot` number of observations that's going to call like this which is defined as this

which is 10 but of course that can be changed so that's where the variables that are set in the config file come into play so the header sort of introduces the um the background libraries and resources that need to be imported um a few parameters are defined so that you can have like a base config set and then you can also just um locally define some variables like the number of initial ants is um you can define variables that are in the config file or you can add some new variables in and then um in the length of the range of this runs a routine the routine is going to be defined better in and stop py but it basically says we're going to run for that many steps for that many starting ants for that many maximum ants we're gonna save it we're gonna switch the location of the and then we're gonna just print some summary descriptors like the number of round trips save that a file so that's like sort of the interface wrapper for running a simulation and then outputting some of the descriptive statistics and then also saving that file now ants.py is 353 lines long so um it's a little bit longer but there are some parts that we could i think walk through the first is that like the ant is a class of object and it has um a few routines it has its initiation routine which is where its x and y position for example are given the starting location and then there's updating a forward and a backwards step there's the class of the environment so this is kind of like if you do net logo modeling agent-based modeling you know you have like the turtle like the agent and then the environment so here we're defining the agent and we're defining the environment and the relationship between the two of them is really whether we have um a stigmaric type multi-agent system or not so you could have the environment influencing the behavior of agents like the sunshine is influencing the behavior of the ants but then the ants don't modify the sunshine so that wouldn't be stigmaric but if there's a possibility for the agent to modify the environment and for the environment to modify the probability of agent behaviors there's some sort of bi-directional relationship then it is stigmaric basically some of the functions that are defined for the environment have to do with uh observations are at a given cell whether the food exists at a given cell whether the walls exist at a given cell and then the step forward a step backward relate to the outgoing and the homecoming trip and this is basically just um doing a delta which i think is a which is the action map it's the delta in position and then it checks whether the ant is returning which is like a gonna be i believe on the step backwards side versus the ant going forward decay is a function that's also inside of environments because decay has to do with the decay of the pheromone at that cell and then the plotting has to do with just saving um the file so that's some of this like uh the gif and the save fig another object type that's defined and this is really where there was the main carry over from the previous models is the markov decision process

so here just like we've talked about in various contexts like the a b c and d are all defined and initiated here those are um used as you'd expect and then the mdp needs to have these functions that relate to setting and resetting certain variables and taking a step and then i love this line 210 action inference like this is sort of the kernel is where it's happening here's the negative expected free energy and then there's a calculation of the negative expected energy of the uh negative utility calculations divided by the scale factor pretty interesting and i don't think we experimented with modifying this scale factor and then um so can i just interrupt you what does that scale factor do really it's mentioned only two times in here i'm not exactly sure what it does here i think that uh when you're comparing relative policies as long as you're dividing them all by the same scale factor it shouldn't change the relative preference for different policies if they're all being scaled down by the same level i'm not sure why the calculation exists this way yeah that i mean it makes sense that they're all like whatever you scaled by something maybe it's just to make the numbers more manageable who knows yes but we'll let's definitely think about it yeah we could get more explanation from alec the uh negative expected free energies get so okay well okay one possible option would be this is like over so we know that the observations are discretized between one and ten so if we divide by 10 maybe we're going from like a 1 to 10 scale to a 0 to 1 scale which then lets us interpret more cleanly as like probabilities um the action ends up being selected based upon the soft max of the negative expected free energies so instead of like you know eight one and one if you do 0.8 0.1 and 0.1 it's really easy to then say okay right 80 likelihood of choosing this action here's a colony level function related to an ant being initiated and just appearing on the map plotting and saving functions and then here's the main a lot of times in the python programs there's like a lot of sort of upstream functions that are going to get called and then main is called and then when main is called that's like where the main program actually happens so this instantiates the environment the ants and the paths it for the number of ants in that are to be specified that many ants are created with reset mdps all of the image files are set to zero the descriptive statistics like how many completed trips and the ant locations are reset is the time point time step based simulation so here for each ant in the ants that exist there's going to be some calculations this is related to the distance calculation here's adding the ants here's switching the food and then here is um checking for each ant trajectory whether it's returning or completed and then saving the file again i think there could be a lot more organization and commenting to sort of understand what these different pieces do but yeah i believe that's some of the main pieces nice are we gonna move to the figures now sure i can

actually try to download the videos and i can play the videos oh yeah that's great so yeah i thought we were going to go switch back and forth from the code to the okay so we'll go to the paper supplemental data let's see if we can play it in the stream directly will it play oh great it will play okay so here is like the ants you'll you won't see it blue but um diffusing around the red ones are going out and the blue ones are coming home so this is just as if you were um watching the colony with video um video two i think we're gonna add in the pheromone density so here the brighter colors are the pheromone density and like there like one ant made at home but then it didn't here comes another one it didn't um sort of lock on but now we can see there's a light ridge it's locked on to the trail and it's going to reinforce the trail it seems like the food has switched because now we're seeing some pheromone getting deposited on the left side and then there's not as much deposition on the right side but this is what really happens in ants like if there's high density of pheromone on that top an ants are spending a lot of their time there especially if they're carrying food they might actually be reinforcing that part so some of the videos are described their similarities and differences in the text but i think the interesting one to look at would be i think six and seven so here there's um here there's a preference for high pheromone density i believe it's seven where the ants can lay down pheromone but they don't have a preference for it and so it ends up getting laid down but then it doesn't get reinforced because they um they just don't prefer it so they act randomly with respect to pheromone but these are just like fun simulations and yeah a total obvious next step an important one would be to explore what sets of parameters certain outcomes occur within like when do they always lock in when do they fail to lock but yeah good supplemental info awesome yeah so what what is the next obvious um step for you or where would you like to see this simulation go most like what is the the next if you were to you know have unlimited time with alec to say like let's code xyz um what what would you uh dream of first i'd probably go for the annotation and clarification because that's like where coding expertise really comes into play and that helps other people modify and build on the code the most when it were a little more reorganized i think it would make sense where other sensory and action affordances could be plugged in but the piece that's such as like adding another pheromone so instead of just having like you know pheromone one through ten density like make it specific pheromone sent one and then that would allow us to go okay now let's just copy and paste pheromone sent two and then we could have um more flavors so well so what what would you make the pheromone scent to do would you make it like fairmont said one drives you to the food and if everyone send two drives you home or would you just make different decay rates

or what would you do with the different fair ones one duality would be like plus and minus if there was the appropriate behavior around using them another one would be fast and slow decay which maps on to the way that ants actually use the sense with like some longer chain hydrocarbons that are longer lasting and some shorter chain ones that are volatile um another interesting cognitive model would be like there's two or three pheromones and then what the inference is on is the ratio and then it could be shown like you know no cheating the ant is only perceiving the ratio of these two things so that would be kind of it's kind of like bimetallism like you tie to the idea of a ratio rather than to just the absolute amount of one so like it could be food could be a little sweet and a little salty or a lot sweet and a lot salty or then the inference of whether it's tasty is like whether it's balanced so then if we're thinking about the ant tasting balances in the world maybe that could help it have an internal manifold that by definition would be simpler than the state space of the pheromone densities out there in the generative so i i heard that there are two types of ants like that there are grease eating ants and sweet eating ants are they just like different like what is that what makes that distinction actually and like does the same ant really eat those things or what's up with that just reminded me of you know like uh uh evil neutral good like the classifications for characters and then what are the major macro molecules what are the nutritious molecules other than vitamins well there's like protein carbohydrates and fats lipids different parts of the colony life cycle might use these different nutrients differently but then what are the resources that could be obtained like sweet that could come from like an extra floral nectary so like a plant that's secreting a honeydew or scale insects and the the carbohydrate-rich fluids that they secrete and ants that love sugar and pies and stuff then there's oil and then there's also like protein like there's meat ants the turtle ants in the in the canopy forests in some rainforests i've heard from many ant researchers like that the most effective traps use human urine and like those ants well that's high nitrogen high protein relatively speaking if you're a tiny guy it's interesting why some species would prefer more or the other if you're putting on more biomass like if you're making larger colonies you probably need a lot more protein and lipids whereas if you have longer lived workers then you don't need as much lipid and protein but you might use a lot of carbohydrates so that's sort of the fun way to think about the workers like a car like you need the metal to make the car but then you only need gas it's like you need the protein and lipids to make the ants but then afterwards you really use more even though they probably also digest other nutrients but i think the the piece that um to connect would be like find some student or person who loves meta-analysis it could be us or it could be someone else and establish like

which data sets are already there what's a recent impactful and publication that already has done a lot of the meta analytic work of at least compiling data sets and there are several ants databases that have compiled behavioral data morphological data ecological data genomic data how could that data format play into model what are we going to put in and then i think uh a good step would just be to show like what does the octave inference ant model explain predict suggest other first principles approaches like reinforcement learning might not perceive very nice yeah so my question about the different types of food that ants eat was was linked to different pheromone like types or ratios or concentrations like if i have a high preference for lipids but i'll eat sugar like you know maybe my ratio preference is different like this is another way to like enable different like types of ants or just to give them different preferences over the food that they eat and then where like if we initiate the same kind of agent and the same kind of food did they all like proportionately distribute or do some carbohydrate preferring ants go to lipids instead of lipid preferring antibiotic carbohydrates i just wonder and like does it have to do with proximity or the number of ants that were there before or i think that all of that is is interesting and kind of relates to like this multi-goal aspect of active inference it's always really kind of difficult to i mean it's easy when we think about and get food right like so the active inference model is as easy it's manageable is what it is so it becomes computationally tractable and like mathematically tractable when it's a single type of agent with a single goal or even like two different types of agents with two different types of goals that are overlapping i mean so you can see just how but it exponentially expands when you have this kind of multi-goal thing and like some goals are like more pressing than others right like my immediate needs have to be met for me to like you know desire to go to college like i have to have food water shelter clothing etc and so these kinds of multi-goal and like prioritizing goal setups are um really interesting and a great way i think to start to modeling real us complex systems that have multiple conflicting sometimes conflicting goals another setting to transpose and explore that where there's a ton of data would be first make active inference with a honeybees so there they don't use stigma g um on the trail because they're on the wing they're flying but in the nest you could have like us in the hive as it were you could have a special dance floor and so that is where maybe like a active inference communication model could come into play that this would be less on the stigmary side but more on like the multi-agent distributed communication side so flip out the trail um modification with a symbolic waggle dance whether it actually does a waggle dance or whether we just say okay there was a waggle dance that occurred and it had this precision and

these estimates honeybees are well studied for making the trade-offs between foraging for nectar carbohydrate pollen which has like lipid and protein and water a lot of work on like is it that um a hundred percent of the variance in that foraging outcome being different is about the needs of the colony and all workers respond to the needs of the colony is it that the workers have their own nest mate level preferences and then the colony outcome just is what it is or is it somewhere in between where there's preferences that differ between workers maybe as a function of their genetic backgrounds but then the colony outcomes are also in feedback with them so that could be um a case and i think this kind of takes us back to um to slide eight which is like when we have the systems lined up and we understand their similarities and we'll be able to transpose across different systems in a really interesting way and then ask um it's kind of like biomimicry on behavioral systems is this more like a honeybee situation where people are exchanging information and then going their own way or is this more like a trail stigmergy situation so like there's going to be some early work transposing into these major different domains like again this was the first paper that transposed active inference and included stigma g although that had been discussed by axel constant and others regarding ecology it had always been qualitatively a part of the discussion that niche modification was relevant and that agents were able to be in feedback with their surroundings and then it went from the natural language through the formalism to the code and then now it can feed back change how we think and write and speak and then we'll transpose into a few of these major domains start to sub-specify into situations and it'll be very cool so it's super interesting about like the um communication versus like the trail like which of these like and i just wonder what you think like as humans which are we more like are we more like bees or are we more like ants like do we communicate and then go on our separate way or do we just like follow in a row where do you think we're at in the b ant trajectory it makes me think of a drive that we went on this weekend and um there was the street signs and then there was a big field and then there was the pleasure path that was cutting the hypotenuse across the field so it was like a big block with an empty field in it and so people were walking and so they were making that trail and so there was this juxtaposition between the um societally imposed and scaffold infrastructure like the streets and the symbolic signs you know the stop sign here and this is where this street and that street intersect and then there's the people who were moving on that landscape and then even if you don't see any other people at that moment it's their stigmergy with the trail and so it is an interesting question and it's almost like we have symbolic rules and scaffolds like law and norms and culture that are even though there's ways to think about

those as stigma energy themselves for the agent level that's more like a symbolic communication it's like the ant stigma g is like blurring the lines that the symbolism provides for us i think there's um there's some components of both plus other insects maybe not insects too but that's quite contentious so that that brings me back to like the concept of of intergenerational sigmoidy but but really like maybe my question was not like phrased correctly like sure we leave some stigmatic traces as humans for like the the next generations but but also like we leave like sigmoidy for strangers right like like so i wonder if like and i know that the ant colonies like the pheromones are different and the different species of ant they're different and so i wonder um maybe is it like the fact that total strangers can go you know searching for um collective intelligence or collective behavior or something like that on the internet and come to the same result or certain like total strangers see the stop sign and know that there's some sort of behavioral norm that's supposed to be enacted just by seeing the signs in the street and so we leave this like you know we do this externalization of of cognition a lot as humans and so this is what i want to see in the animal kingdom like like i want to see traces of like can strange animals say like oh i see a rock stack i know the trail goes this way right like like does that happen in the animal kingdom or is that what's maybe uniquely human good question and i think this is why we can study these systems and gain insight without naively copying their strategies or thinking that we must be them or they must be us because our whatever it is that we're doing it's at the intersection of like stigma g and cumulative culture modification and symbolism so in a way the ant and the bee foraging paradigms not no hashtag not all ants because some don't do pheromone trails etc but one extreme is like the syntax of the foraging trail and there's no symbolic communication it's just stigma g and then the other extreme is no stigmatic modification there's no trace left in the environment at all there's just transient symbolic communication so it's this continuum from the transient symbolic like i'm going to pass you a message and then you're going to be off on your way the stigmatic and often sub symbolic like the path crossing that field it isn't directly a symbol saying walk on me it isn't in affordance we perceive it as an ecological affordance but it doesn't have it that's not what it symbolizes that's what it is and that's how we enact it um so like we could have some types of active inference models that highlight that um transient among agents meaning based communication versus models that um don't need to take on any grandiose hermeneutic communication and what does communication mean and how does thinking through other minds happen we just look to the ground and about how agents respond to their niche which is a combination of biotic and abiotic fun times yeah just any final

thoughts or or any anything else for me i've given all my all my thoughts out i think oh me too over the years but thanks for um pushing for this paper for 29 and uh 30 is gonna be hot on the heels slash abdomen of this paper because i think we're going to be able to operationalize a lot of the discussions that we had about extended cognition i know it's also going to bring back what we talked about with the definitions of a self and individuality another one in the books on to our third decade papers on the live stream and uh yeah i hope people found it interesting as with other papers except this is one where hopefully they can get in touch with the authors and understand that it's a totally open area and a collaborative one too awesome well thanks for giving us your time and and always giving us your time thanks youtube luke so um i'll flip to the closing slide that's what we can still think about awesome see everyone around next time